

FASALRAKSHAK फसल रक्षक

Deadline-Aware PMFBY Loss Intimation Agent

AGENTIC AI

WHATSAPP + VOICE

HINDI / HINGLISH

Team: Ctrl C + Ctrl V

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1. TARGET USER & CONTEXT

Profile: Small/marginal farmers enrolled in PMFBY via Kisan Credit Card.

Pilot Scope: Narrow single-district pilot (Sehore, Madhya Pradesh).

Context: Owns or shares a smartphone, comfortable with WhatsApp voice notes. Uncomfortable with complex government portals, English forms, or locating precise policy numbers. Often unaware of the strict timeline until it is too late.

2. THE WORKFLOW PROBLEM

Under PMFBY, farmers must file a loss intimation **within 72 hours** of crop damage.

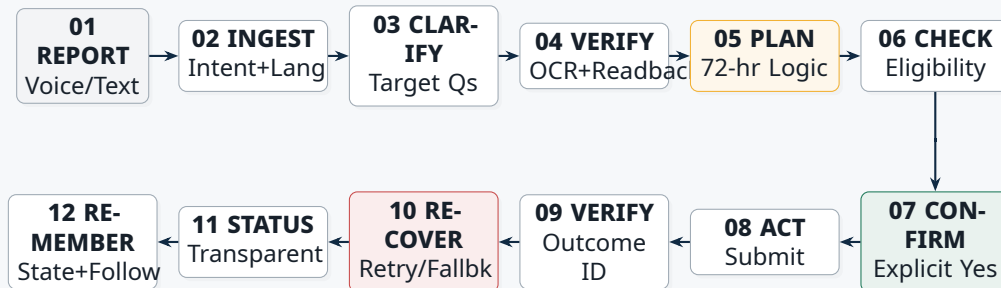
Many valid claims fail not because of fraud, but because of a **workflow gap**: toll-free lines drop, apps require unavailable data, or details are mismatched. The farmer knows the crop is ruined, but the static reporting process acts as a barrier, causing the 72-hour deadline to expire.

3. TASK DEFINITION: WHAT FASALRAKSHAK ACTUALLY COMPLETES

Convert a farmer's unstructured spoken/WhatsApp report of crop damage into a **complete, correctly formatted loss intimation** submitted through an available channel, safely inside the 72-hour window.

Definition of "Done": A valid intimation tracking ID is issued and confirmed back to the farmer **OR** a fully verified, CSC-ready data packet is generated if all remote channels catastrophically fail.

4. CORE AGENTIC WORKFLOW



5. WHY AGENTIC AI?

Traditional Chatbot

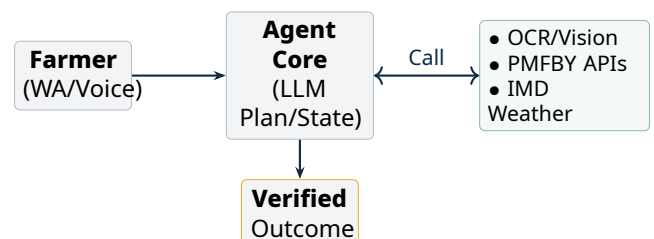
Scripted Q&A → Retrieve Answer → Stop.
Fails instantly on out-of-order input, cannot calculate deadlines, cannot take actions, and drops the user if a government API times out.

FasalRakshak

Understand → Plan → Use Tools → Act → Verify → Recover.

Reasons about missing data, intelligently switches channels upon failure, and persists state across a multi-day lifecycle to guarantee task resolution.

6. PROPOSED ARCHITECTURE



7. TOOLS & INTEGRATIONS (FEASIBILITY)

Component Integration	Status for Demo	Role & Real-World Path
WhatsApp Business API	REAL	Primary farmer interface. Free-tier sandbox used for Round 2 prototype.
ASR (Hindi/Hinglish)	REAL	Bhashini API or Whisper to process voice notes into text.
Agent Planner / Tool Caller	REAL	Core logic (e.g., Claude/GPT API) driving the decision loop.
Vision / Land Record OCR	REAL	Extracts Khasra/Survey numbers from photos. <i>Human verification enforced.</i>
PMFBY Enrollment Lookup	MOCK	Real NCIP is not public API accessible. Mocked faithfully against PMFBY schema.
Intimation Submission	SIMULATED	Simulated portal endpoints to demonstrate failure-recovery and deadline logic.
Weather Corroboration	OPEN DATA	IMD open data API to verify storm/rain dates against farmer's claim.
State Persistence	REAL	SQLite/Postgres to allow the agent to remember context over 72 hours.

8. DATA INPUTS & CORE AGENTIC RECOVERY MATRIX

Data Flow Strategy: Voice/Text + Photos → Extract Fields → Confidence Check → **Explicit Read-Back** → Verified State.

The agent never silently auto-corrects or guesses critical fields. Unstructured, messy input is resolved via the following recovery logic:

Failure Scenario	Agent Detection	Planning Decision	Recovery Action	Final State
1. Blurry Document Photo	Vision API confidence below threshold.	Do not hallucinate a survey number.	Prompt farmer for retake or accept verbal fallback.	Field correctly populated via trusted source.
2. API Channel Timeout	Simulated portal returns 503 / timeout.	Retry with backoff (3x). If failed, urgency requires fallback.	Switch to SMS fallback & generate CSC packet.	Intimation recorded; farmer notified of alternate path.
3. Enrollment Mismatch	Lookup returns NULL for farmer ID.	Do not fabricate a successful claim.	Re-ask details. If still failed, honest redirect to bank.	No false success reported; transparent handoff.
4. Contradictory Input	Damage date vs Weather data mismatch.	Flag inconsistency, do not auto-correct.	Ask targeted clarifying question to farmer.	Data corrected and verified prior to submission.

9. USER SAFETY & CONSENT

Consequential Action Gate: The final filing step always requires explicit consent before execution:
"आपके लिए फसल नुकसान की सूचना दर्ज करने वाला हूँ। क्या मैं अभी सबमिट कर दूँ?"

No Fabricated Success: The agent strictly delineates what it *knows*, what it *retrieved*, and what it *has completed*.

Privacy: Minimal PII retention; Aadhaar/Bank details processed only for the session and redacted in logs.

11. TECHNICAL RISK & MITIGATION

Risk: Hindi/Hinglish ASR and Khasra OCR reliability fluctuates heavily based on farmer accent and photo quality.

Agentic Mitigation:

- Confidence Thresholds:** Low confidence triggers a clarifying follow-up, not a hallucination.
- Read-back Protocol:** Every extracted field is read back to the farmer for confirmation.
- Structured Fallbacks:** WhatsApp quick-reply buttons are injected to reduce open-ended ASR dependency for critical dates and crops.

10. EVALUATION PLAN

Metrics tested against 20 scripted real-world scenarios:

- Completion Rate:** TARGET > 85% (Valid intimation generated within 72hr rule).
- Field Extraction Acc:** TARGET > 85% (Voice+OCR).
- Failure Recovery:** TARGET > 70% (Successfully bypassing injected API timeouts).
- Time-to-Intimation:** Median < 15 minutes.

12. FEASIBILITY FOR PROTOTYPE

Build timeline is highly credible for Round 2. We avoid sprawling scope by focusing on a **single-district pilot**. The core stack relies on accessible tools (WhatsApp API, LLM tool-calling, Tesseract/Vision). Mocked components (NCIP) are explicitly disclosed and modeled faithfully on documented schemas, proving the logic without requiring impossible government access.

FasalRakshak does not just tell a farmer what to do. It perceives, plans, acts, verifies, and recovers until the task reaches a trustworthy, verifiable outcome.